MILITARY SPECIFICATION

INERT FILLERS FOR USE IN AMMUNITION

1. SCOPE

1.1 **Scope.** This specification covers the types of inert fillers used in loading of standard items of ammunition.

1.2 **Classification.** Inert fillers shall be of the following compositions:

Type I — Mixture of calcined gypsum, perlite and water.

Type II — Mixture of the glyceride of 12 hydroxy stearic acid, rosin and dead burned gypsum with or without ferric oxide.

Type III — Mixture of aluminum ammonium sulfate, cellulose or barium sulfate, copper sulfate and a corrosion inhibitor.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

LLL-R-626 — Rosin.

O-A-401 — Alum, Ammonia (Aluminium, Ammonium Sulfate), Technical.

O-C-828 — Cupric Sulfate, Pentahydrate, Technical.

PPP-D-723 — Drums, Fiber.

MILITARY

MIL-G-2550 — General Specifications for Ammunition except Small Arms Ammunition.

MIL-P-13900 — Perlite.

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-129 — Marking for shipment and storage.

MIL-STD-650 — Explosive: Sampling, Inspection and Testing.

MIL-STD-1167 — Ammunition Data Cards.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

PURCHASE DESCRIPTIONS (see 6.3)

PA-PD-290 — Gypsum, Calcined.

FSC 1395

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PA-PD-656 — Iron oxide, Ferric.

PA-PD-721 — Glyceride of 12-Hydroxy stearic acid.

PA-PD-722 — Gypsum, dead burned.

PA-PD-2765 — Cellulose, powdered.

PA-PD-2766 — Barium sulfate.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Inert filler composition.* The material in the inert filler compositions shall be in accordance with applicable specifications cited in 3.1.2. The exact parts by weight indicated in 3.1.2 may be varied by the manufacturer; however, no ingredients other than those specified shall be utilized and no ingredients shall be omitted.

3.1.2 *Types.*

Type I

Constituent material	Conforming to	Parts by weight
Calcined gypsum	PA-PD-290	96 (see 6.5)
Perlite	MIL-P-12800	3-6 (see 6.5)
Water		30 (see 6.5)
Sodium Citrate		(See 6.6)

Type II

Constituent material	Conforming to	Parts by weight
Glyceride of 12-hydroxy stearic acid	PA-PD-721	35±1
Eosin	LLL-R-626, Type I, Grade FF	5±1
Dead burned gypsum	PA-PD-722	(See 6.7)
Ferric oxide	PA-PD-656, Type II	(See 6.7)

Type III

Constituent material	Conforming to	Parts by weight
Alum, Ammonia	O-A-401 (see 6.8)	100
Powdered cellulose	PA-PD-2765	(See 6.11)
Barium sulfate	PA-PD2766	(See 6.11)
Acetone inhibitor No. 11	(see 6.9)	0.75
Cupric sulfate	O-C-828 (see 6.10)	1
Water		4

3.2 Specific Gravity. The specific gravity of the finished composition shall vary no more than ± 0.04 from that specified in the drawing or in the contract.

3.3 Workmanship. The composition shall be free of dirt, chips, grease, and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 *Contractor quality assurance system.* If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment), and provisions for control and disposition of nonconforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedures of this specification shall apply. The contractor shall notify the Government, of and obtain approval for, any changes to the written procedure effects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all requirements and provisions of the approved current written description of the filler.

(b) Specification number and date, together with an identification and date of changes.

(c) Certificates of analysis on all material used directly by the contractor when such material is controlled by Government specification, shall be made available upon request by the contracting officer.

(d) Quantity of product in the lot.

(e) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, (see 4.1.1) this detail specification, and other contractual documents as a guide, the Government shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with (a) or (b) as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance that all quality provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and, depending on the nature of the deviation, may consist of lot rejection, screening, re-sampling, re-instruction or the supplier's employees, or other appropriate action:

(a) Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

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(b) Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 *Surveillance.* Surveillance shall include, but is not limited to:

(a) Observation of procedures concerning lot formation and identification.

(b) Observation of sampling procedures and application of acceptance criteria.

(c) Determinations that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

(d) Review of procedures for control and disposition of nonconforming material.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of one or more batches of the inert filler produced by one manufacturer in accordance with the same specification, or same specification revision under one continuous set of operating conditions. Each batch shall consist of that quantity of the inert filler that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235 when applicable).

4.2.2 *Examination.* Sampling plans and procedures for the following specification of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans, in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect.

4.2.2.1 *Drums, prior to filling (Polyethylene liner).*

Categories	Defects	Method of inspection	Code No. (see 6.12)
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Liner cut, torn or punctured	Visual	01001
Minor: None defined.			

4.2.2.2 *Drums, prior to closing (Polyethylene liner).*

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Liner not completely sealed	Visual	02001
Minor: None defined.			

4.2.2.3 *Drum, sealed (fiber).*

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Weight of contents	scale	03001
102.	Closure incomplete or damaged to the extent that contents sift out	Visual	03002
Minor:	AQL 0.65 percent		
201.	Markings misleading or unidentifiable	Visual	03003

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4.2.3 *Testing.* A 10- to 15-gram specimen of the cast inert filler shall be selected from each batch produced and tested in accordance with the test specified in 4.3.1. If the sample fails to comply with the specific gravity specified on the drawing or in the contract, the batch shall be rejected.

4.3 Test methods.

4.3.1 *Specify gravity.* Code No. 05001. Determine the specific gravity of the cast inert fillers in accordance with Method 203.1 described in Standard MIL–STD–650.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Drums.* Unless otherwise specified in the contract or purchase order, the inert filler shall be packed in nonreturnable fiber drums in accordance with Specification PPP–D–723 (Type III, Grade A). The drums shall be finished with a nominal 0.004 inch thick polyethylene liner, properly heat sealed or otherwise closed to afford protection to the contents equivalent to that given by a heat seal closure.

5.1.2 *Level C.* The inert filler shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules and Container Specifications for rail shipments or National Motor Freight Rules and Container Specifications for truck shipment as applicable.

5.2 Marking. In addition to any special marking required by the contract or purchase order, containers shall be marked in accordance with MIL–STD–129. Marking shall include, but not limited to, the following information:

(a) Manufacturers name.

(b) Product designation.

(c) Lot number.

(d) Date of manufacture.

(e) Number of this specification.

6. NOTES

6.1 Ordering data. Procurement documents should specify the:

(a) Title, number, and date of specification.

(b) Type required.

(c) Ammunition data cards shall be prepared in accordance with Standard MIL–STD–1167.

6.2 Intended use. The materials described in this specification are intended to be used as inert fillers in place of explosives in artillery shells as well as other standard items of ammunition.

6.2.1 Type I filler simulates such high explosives as composition B and 80/20 Tritonal.

6.2.2 Type II filler simulates such high explosives as TNT, Composition B, 52/48 Picratol, 75/25 Cyclotol, 80/20 Tritonal, HBX–1, HBX–3 and HBX–6.

6.2.3 Type III filler simulates explosives having specific gravity from 1.5 to 1.95.

6.3 PA–PD documents. These documents are purchase descriptions prepared by Picatinny Arsenal, Dover, N. J.

6.4 Compatibility with explosives. Type III filler is compatible with TNT, RDX, and HMX.

6.5 The proportions given are approximate. These proportions have to be varied to produce an inert filler of the specific gravity prescribed on applicable drawing.

6.6 The addition of not more than 0.01 grams of Sodium Citrate for each pound of

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calcined gypsum in Type I filler is permissible to increase the time of pourability.

6.7 The parts by weight of dead burned gypsum and ferric oxide shall be varied in the proportion necessary to produce a filler of the weight prescribed on the applicable drawing. To assist in determining the parts by weight of dead burned gypsum to be replaced by ferric oxide necessary to meet the weight requirement specified on the applicable drawing, reference may be made to figure 1. This figure indicates the approximate specific gravity produced by the nominal combination of constituents.

TABLE 1. *Relationship between composition and specific gravity*
Type III Filler

Composition, %			Specific gravity
Alum	Cellulose	$BaSO_4$	
100	.....	.....	1.65
95	5	.....	1.58
90	10	.....	1.50
90	.....	10	1.71
85	.....	15	1.77
80	.....	20	1.85
75	.....	25	1.95

6.8 A rice size has been found more suitable than the nut size described in specification.

6.9 Actane inhibitor, No. 11 is a proprietary item and may be obtained from Enthone Inc., 442 Elm Street, New Haven, Conn.

6.10 Powdered crystals shall be used rather than the particle size given in specification.

6.11 The parts by weight of cellulose and barium sulfate shall be varied in the proportion necessary to produce a filler of the weight prescribed on the applicable drawing. To meet the weight requirement specified on the drawing, reference may be made to figure II and table I. Picatinny Arsenal's Technical Report No. 3276, "Development of a Dimensionally Stable Inert Filler for Ammunition" describes in detail the method of manufacture of Type III inert filler.

6.12 **Inspection code numbers.** The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

Custodian:

Army—MU

Preparing activity:

Project No. 1395-A006

Army—MU

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SPECIFIC GRAVITY

COMPOSITION RELATIONSHIP FOR TYPE II FILLER

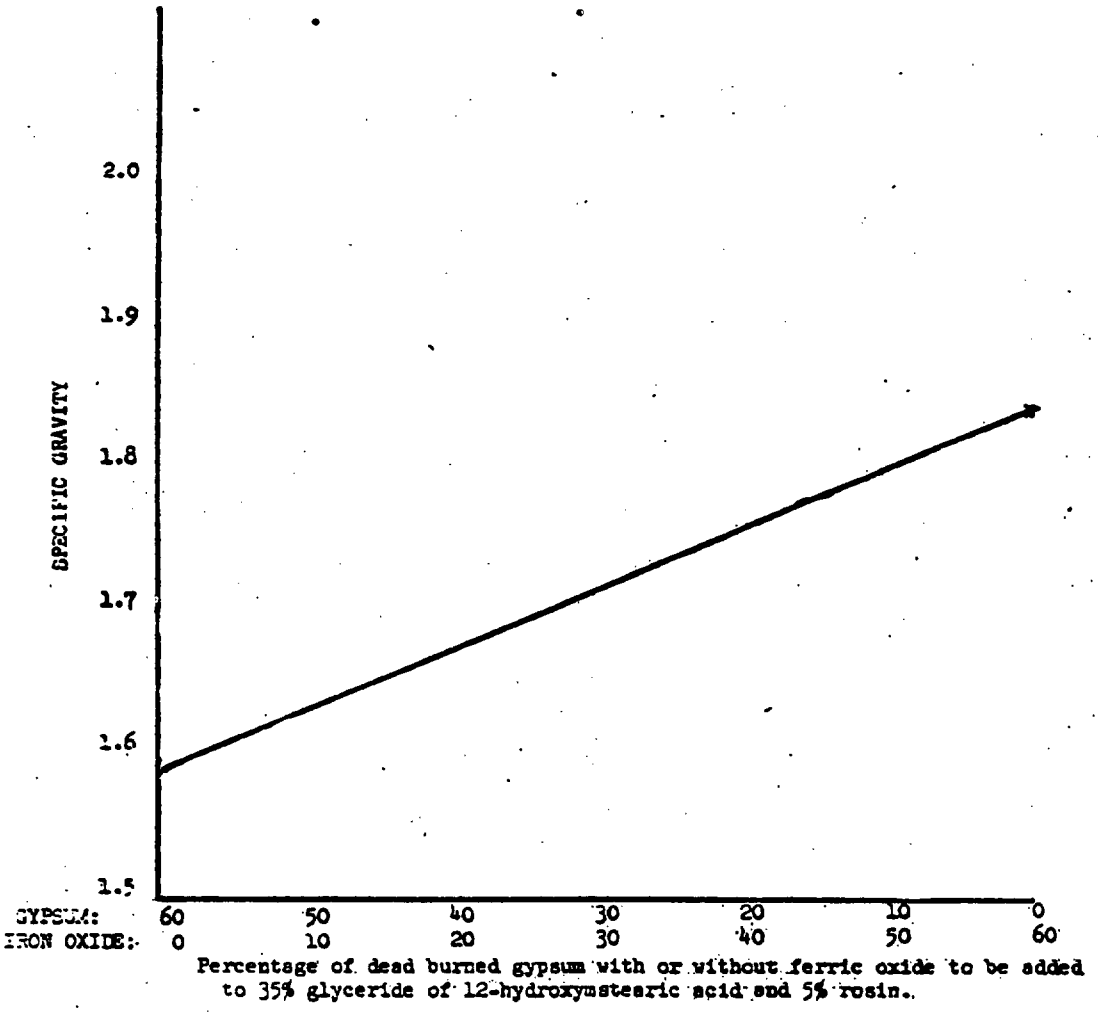


FIGURE I.

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SPECIFIC GRAVITY

COMPOSITION RELATIONSHIP FOR TYPE III Filler

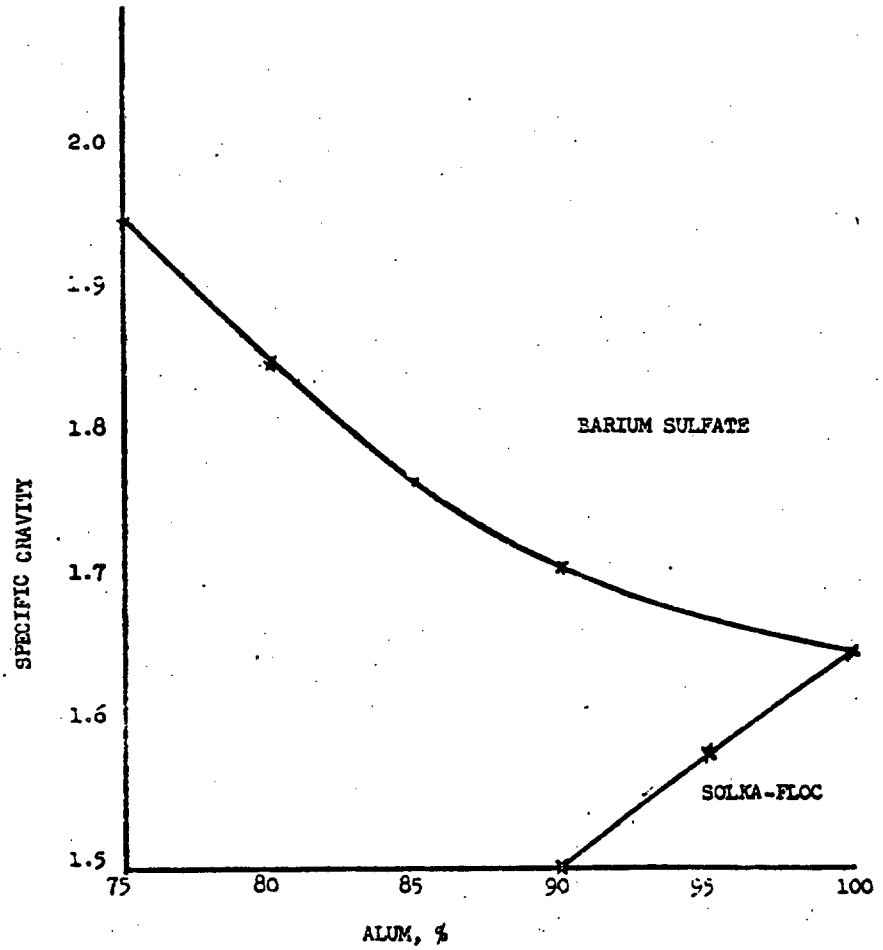


FIGURE II.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426 APR 63 REPLACES NAVSHIPS FORM 1863, WHICH IS OBSOLETE.

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